WAFIY ALMAS TSAMAR

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SUMMARY

Fresh graduate in Nanotechnology Engineering, Universitas Airlangga (GPA 3.52), and **Lean Six Sigma Green Belt** with about one year of research experience at BRIN in **lithium-ion batteries (NMC, LCO, LMO, LFP)**, battery recycling using **hydrometallurgy & pyrometallurgy**, and **hydrogen technology**. Trained in **PFD/P&ID/BFD reading**, **Unit Operations in Chemical Engineering**, **SOPs**, **HIRADC/JSA**, and **ISO 9001/ISO/IEC 17025**, with skills in **process data collection, statistical analysis, root cause analysis, and DMAIC**. Demonstrated **leadership** as **Chairperson** of a university-level event, **Head of Scientific Department**, and organizing committee member for three hydrogen FGDs at BRIN with 100+ energy companies. Eager to **learn and contribute** as a **Process Engineering Intern**, supporting process calculations, equipment performance tests, and quality improvement in **EPC and battery manufacturing** environments.

EDUCATION

Bachelor of Engineering (S.T.) - Nanotechnology Engineering | Universitas Airlangga *Aug 2022 - Jul 2026*

- Faculty of Advanced Technology and Multidisciplinary; achieved GPA: 3.52/4.0.

WORK EXPERIENCE

Badan Riset dan Inovasi Nasional (BRIN) | South Tangerang, Indonesia

Research Intern Merdeka Belajar Kampus Merdeka (MBKM) & Event Committee Member *Aug 2025 - Jul 2026*

- Conducted research on lithium-ion batteries, battery recycling (hydrometallurgy & pyrometallurgy), and hydrogen technology, including fuel cells, electrolyzers, and catalytic materials.
- Served as an organizing committee member for three hydrogen technology focus group discussions (FGDs) activities with Kementerian ESDM EBTKE and directors from 100+ companies.
- Served as exhibitor/booth staff representing BRIN and Indonesia Fuel Cell and Hydrogen Energy (IFHE) at an event at ICE BSD, Tangerang.
- Developed experience in experimental planning, laboratory work, data interpretation, and scientific communication.

Amrillah Renewable Energy Research Group (AmrilLab) | Surabaya, Indonesia

Research Assistant *Sep 2024 - Apr 2025*

- Synthesized and characterized semiconductor nanomaterials for thin-film solar cells and photocatalysis, including CZTS, Ni-doped CZTS, BiOBr, and Al-doped ZnO.
- Evaluated solar cell performance using a solar simulator, including PCE, FF, Voc, and J-V curve analysis.
- Supported material development through systematic testing, data analysis, and research documentation.
- Co-authored the review article “Kirigami and Origami-Inspired Structures for (Flexible) Batteries and Supercapacitors” (manuscript under review).
- Co-authored the review article “Crafting a Solar Cell Device Integrated Textile” (manuscript under review).

ORGANIZATIONAL EXPERIENCE

Airlangga Student Research Group of Nanotechnology (ARGON)

Developed ARGON & Researcher at sensors & electronics division *Feb 2025 - Dec 2025*

- Developed ARGON’s organizational grand design and served as a researcher (sensors & electronics division)
- Contributed to electronics and sensor-related activities within the student research group.
- Concept Developer | Led conceptual design and coordination of a hydrogen-based electrolyzer system.
- Coordinated project development within the student research group.

Nanotechnology Engineering Student Association (HIMANO)

Head of Scientific Department *Feb 2024 - Dec 2025*

- Led the scientific department and coordinated research- and academic-oriented activities.
- Worked with organizational members to plan and execute departmental programs.

Airlangga Nanotechnology Week as part of Airnology 3.0

Chairperson *Feb 2024 - Sep 2024*

- Led the organizing committee and coordinated cross-functional teams throughout event planning and execution.
- Oversaw event operations, sponsorship and funding strategies, and coordination across event functions.
- Contributed to drafting Memoranda of Understanding (MoUs) for event partnerships and coordinated collaboration with external stakeholders.
- Ensured effective collaboration and alignment to achieve event objectives.

Tutoring Class Program

Person in Charge (PIC) *May 2023 - Jun 2023*

- Coordinated the implementation of a tutoring class program.

Community Service & Outreach

- Belajar Bersama Komunitas (BBK) | Made II, Sambikerep, Surabaya | Dec 2024 - Jan 2025 (Taught mathematics to students and delivered basic economics education to local residents).
- Organic Waste to Renewable Energy Outreach | Sutorejo, Surabaya | Sep 2023 (Promoted zero-waste lifestyles through community education on organic waste conversion into biogas).
- BEM FTMM x ARGON Community Outreach | Gempol, Pasuruan | Jun 2025 (Delivered practical education on nanochitosan fertilizer production and formulation, from raw materials to a farmer ready product, supported by SDGs UNAIR funding).

ACHIEVEMENTS

- Finalist – "Kompetisi Ilmiah Mahasiswa (KIM) 2025, Universitas Airlangga",
- Incentive Recipient – "Program Kreativitas Mahasiswa (PKM-GFT) 2024"
- Finalist – "International Youth Summit 2024, Kuala Lumpur, Malaysia".

CERTIFICATIONS & TRAINING (View Certificate)

- **Lean Six Sigma Green Belt.**
- How to Read P&ID, PFD & BFD used in Process Plant
- Unit Operations in Chemical Engineering.
- ISO 9001:2026 - Quality Management System.
- ISO/IEC 17025 - Laboratory Management Systems.
- Job Safety Analysis (JSA) dan Hazard Identification, Risk assessment and Determining Control (HIRADC) Awareness,
- Basic Occupational Safety and Health (K3) Awareness.
- Awareness Personal Protective Equipment (PPE/APD) Awareness,
- Supply Chain- MPS-RCCP-MRP-DRP-BOM
- Measuring Manufacturing Effectiveness-OEE, OOE, and TEEP
- English Language Proficiency Test (ELPT),

ADDITIONAL

Data & Software: Microsoft Excel, OriginPro, Gaussian, DWISM, BIOVIA Material Studio, HyperChem, ImageJ, Python, C++, SOLIDWORKS.

Technical Knowledge: Lithium-ion Battery Technology, Battery Recycling, Hydrometallurgy, Pyrometallurgy, Hydrogen Technology and Solar Cells.

Professional Skills: Teamwork, Communication, Public Speaking, Time Management, Analytical Thinking, Problem Solving, project Coordination, Research & Development, Scientific Writing, High Learning Motivation, Leadership.